CrackID: A Benchmark for Crack Re-Identification, and the Limits of Appearance Matching

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***Abstract*—Automated building-maintenance systems increasingly convert photographs into tracked records: a defect is photographed, a ticket is opened, and follow-up activity accumulates against that record. Any defect re-photographed across visits therefore needs re-identification — a new photo must match the record for that specific physical defect, or open a new one. Wall cracks are a common and difficult case. Yet despite crack detection being mature, we are not aware of a dataset or benchmark for crack *re-identification*, and the natural approach — crop the region and match embeddings — imports an assumption from person and vehicle re-ID that may not transfer: that the object has stable, locally distinctive appearance. We introduce CRACKID: 140 photographs over 17 walls with 212 annotated crack identities, released with an explicit annotation-quality report rather than as unaudited automatic labels, and with the chance level of every metric stated. Benchmarking ten matchers across classical keypoints, learned correspondence, general vision embeddings, a re-identification architecture, and a purpose-built crack-shape descriptor, we find that on the decision a maintenance system actually makes — *same crack or not*, at a threshold — none exceeds chance by more than 0.054 pairwise F_1, three achieve nothing at all, and at a validation-calibrated threshold nine of the ten span only -0.012 to $+0.012$ around chance while one — the most expensive — holds a real margin of 0.046. Two controls locate the difficulty: giving an embedder the surrounding wall raises its open-set rate from 0.168 to 0.404, but *erasing the crack* from that same view costs almost none of it (0.393) — so what such a method retrieves on is location, not crack appearance. As a geometric reference baseline we register on wall texture with cracks masked *out*, then Chamfer-match projected crack masks under global assignment. It reaches a detection-and-identification rate of 0.768 at 10% false alarm, against 0.429 for the best crop method and 0.000 for a control that exploits its registration coverage without ever examining a crack — so the open-set gap is attributable to matching rather than to partial coverage, while the ranking gap is substantially not.**

***Index Terms*—crack re-identification, dataset, benchmark, building maintenance, structural inspection, homography, geometric alignment, annotation quality**

I. INTRODUCTION

Automated building-maintenance systems increasingly convert photographic observations into tracked records. A defect is photographed, a ticket or issue is opened, and subsequent activity — status changes, follow-up inspection, resolution — accumulates against that record over the defect's lifetime. This creates a re-identification requirement for any defect type that gets re-photographed across visits: a new photo must match the existing record for that specific physical defect, or a new record must be opened, without a human reconciling every submission by hand. Wall cracks are a common and difficult case, since the same crack is routinely re-photographed from a different distance, angle, or lighting condition across separate inspection visits — and, over a long enough gap, the crack itself has grown (Fig. 4 pictures the effect; Sec. VIII-H explains why CRACKID, a single-visit dataset, does not yet measure it).

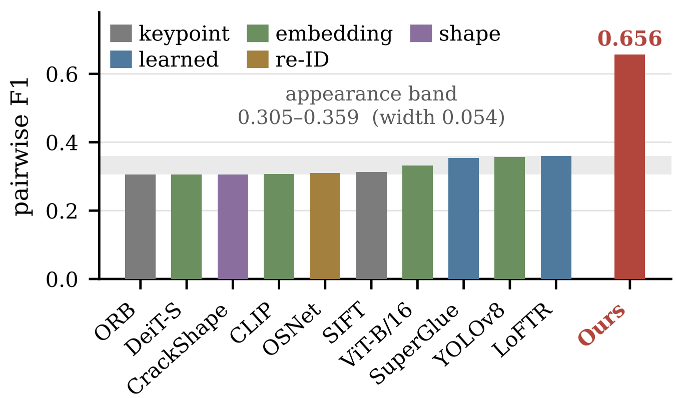

Fig. 1. **The appearance ceiling.** Pairwise F_1 for ten crop-scope matchers spanning five method families, on the CRACKID test split. Every method falls inside a band of width 0.054, irrespective of family, era, or compute budget. The geometric reference baseline (red) leaves the band. Ranking metrics separate these same methods over a 0.30 range; the decision metric does not.

Crack analysis is a mature computer vision task, but a mature *detection* and *segmentation* task: the public datasets are annotated within single images, and the largest consolidations of them — CrackSeg9k's 9k images [1], OmniCrack30k's 30k over twenty sub-datasets [2] — carry no cross-observation identity labels, because the task they serve does not need any. We are not aware of a dataset or benchmark for crack *re-identification*: deciding which recorded crack a new observation corresponds to, or whether it is new (Sec. II-B).

This gap matters because the natural approach — treat it as instance re-identification, crop the reported region, and match embeddings — carries an assumption from person and vehicle re-ID [3], [4] that does not obviously transfer: that the object being matched has stable, locally distinctive appearance. A crack is thin, low-texture, and locally self-similar; whether crop appearance is actually sufficient to re-identify one has, to our knowledge, never been measured.

We introduce CRACKID to make that measurement possible, and to establish where the difficulty of this task actually lies.

Appearance buys almost nothing on the decision. On ranking metrics the ten benchmarked matchers separate as expected — Rank-1 spans 0.352 to 0.654, against a chance level of 0.336. But on pairwise F_1, the metric measuring the decision a maintenance system actually makes (*same crack or not*, at a threshold), the picture is different in kind. That metric has a hard floor: we report the best F_1 over a threshold grid, and the grid contains the degenerate point at which every pair is called positive, which scores $2p/(1+p) = 0.305$ here whatever the scores are. All ten methods land between 0.305 and 0.359 — that is, between 0.000 and 0.054 *above chance* (Fig. 1). ORB, DeiT and the shape descriptor we designed specifically for cracks achieve exactly zero. At a threshold calibrated on held-out validation walls rather than on the test matrix, excess over chance for nine of the ten tight-crop methods spans only -0.012 to $+0.012$ — two of them below chance — and one, LoFTR, holds a real margin of 0.046.

Two controls say where the difficulty is not. Widening the crop until an embedder sees the surrounding wall helps a great deal — OSNet's open-set rate goes $0.168 \rightarrow 0.404$. Then *erasing the crack* from that same widened view costs almost none of it (0.393). A crop+context method is therefore retrieving on the wall behind the crack, not on the crack; and the cue that works, even in that crude implicit form, is *location*.

Building it required solving an annotation problem. Propagating a single click-point identity across a wall's photographs via homography is the only tractable way to label crack instances at this scale, but it fragments a physical crack into multiple identities whenever propagation breaks, requiring a merge step — and a merge that incorrectly joins two distinct cracks silently converts a hard negative into a free positive and inflates every downstream matching metric at once. We treat this as a first-class part of the contribution rather than a footnote: 45% of CRACKID identities were formed by such a merge, and we report that rate, the gap distribution it bridged, and the agreement between merged and pre-merge labellings, instead of presenting the result as ground truth by construction (Sec. III-C).

Contributions.

1) CRACKID, 140 photographs across 17 walls and 212 crack identities, released with an explicit annotation-quality report — merge rate, bridged-gap distribution, chance-corrected agreement against the pre-merge labelling — plus the instrumentation for a human error-rate audit (Sec. III).
2) A ten-method benchmark spanning classical keypoints, learned correspondence, general vision embeddings, a dedicated re-identification architecture, and a purpose-built crack-shape descriptor, measuring how much each buys *over chance* on the decision metric — between 0.000 and 0.054 — together with crop+context and crack-erased controls that localise the difficulty to crack appearance rather than to field of view (Sec. V).
3) A geometric reference baseline — registration on wall texture with cracks masked out, followed by Chamfer-matched crack masks under global assignment — reported against a COVERAGE-ONLY control that exploits its exact registration coverage without examining a crack, so that alignment and matching are separated rather than confounded (Sec. VI, VII).
4) A protocol for this task's failure modes: chance levels reported for every metric because two of them are far from zero, a frame-gap sweep that turns the benchmark's near-duplicate structure into a stated difficulty axis, a blur admission gate set by measured impact on registration yield rather than convention, a viewpoint covariate estimated independently of the scoring pipeline to avoid circularity, coverage-aware metrics that keep unanswerable queries and charge unscorable pairs, and leave-one-wall-out resampling in place of a point estimate on a small identity count (Sec. IV).

II. RELATED WORK

A. Crack datasets, and where they stop

Automated crack analysis is overwhelmingly a detection and segmentation literature [5], [6], and its datasets follow: CrackTree [7], CFD [8], CRACK500 [9], DeepCrack [10], GAPs [11] and their successors annotate cracks *within* single images, and the large consolidations merging them [1], [2] inherit that scope. None carries an identity label linking a crack in one photograph to the same physical crack in another, because segmentation does not need one. They answer *where is there a crack in this image*, never *is this the crack from last time*, and the distinction is not cosmetic: a detector perfect on every visit still yields an incoherent defect history if the visits cannot be linked. Multi-view pipelines aggregate across images of one inspection [12] but establish no instance correspondence, so they cannot support temporal tracking either.

B. Multi-temporal crack monitoring: the nearest neighbour

Structural health monitoring does compare crack images across time, with machinery close to ours. Parente *et al.* [13] track crack width from repeated captures; Harb *et al.* [14] use temporal sequences to improve crack *segmentation*; Sun and Chang [15] perspective-correct handheld time-lapse images so evolution can be measured under field distortion. RANSAC-estimated homography is the standard tool throughout, and it appears in patents too, matching a new crack image against registered ones via SIFT [16].

The task is different in a way that matters. Throughout this work the *identity of the crack is given by construction* — a fixed camera, a marked location, or an operator guided back to a previous viewpoint supplies the correspondence — and the open question is the crack's width or extent. Accordingly the metrics are measurement metrics (width, area, spine-length error) rather than association metrics; the subject is one known crack rather than a wall of candidates; and no dataset is released on which one association method could be compared against another. The patent case is telling: its SIFT-and-homography pipeline exists to guide an operator back to the same camera pose, which *avoids* the re-identification problem rather than solving it.

What is missing is the decision an automated maintenance record requires: given a new photograph of a wall carrying several cracks, which recorded identity does each observation belong to, and which are new? That is an open-set association problem over a gallery, it has no benchmark, and it is what CRACKID supplies.

C. Object re-identification

Re-identification matches an instance across captures [4], and open-set identification — deciding whether a probe is in the gallery at all — is measured by the detection-and-identification rate against false alarm rate that we adopt as our headline metric [17]. Handcrafted local descriptors [18]–[20] are efficient and interpretable but need repeatable, distinctive keypoints — precisely what hairline cracks on flat render fail to provide. Learned correspondence replaces descriptor matching with attention: SuperGlue [21] refines detected keypoints through a graph network and an optimal matching layer, and LoFTR [22] is detector-free and dense. Both are markedly more robust, and both are pairwise, so cost grows with gallery size. Embedding methods map a region to a vector [23]: ViT and DeiT [24], [25], CLIP [26], detector backbones [27], and OSNet [3], which is designed for instance-level discrimination.

Our benchmark evaluates all of these. Our finding is not that any one of them is weak, but that on the decision metric none of them clears chance by more than 0.054 — a result that only becomes visible when the population is broad enough, the metric is the deployable one rather than a ranking statistic, and the chance level is reported beside it. We have not evaluated a current self-supervised backbone such as DINOv2 [28], nor SuperGlue's faster successor LightGlue [29]; both are the natural next entries against this benchmark, and the framework and protocol here are what make adding them cheap.

D. Damage association in inspection pipelines

The closest integrated systems come from vehicle-damage assessment. Maiano *et al.* [30] combine filtering, detection, and OSNet-based damage re-identification for insurance claims. Commercial vehicle inspection platforms rely on controlled capture rigs. In both cases association is posed over cropped damage regions and evaluated where acquisition is curated. Our results suggest the crop scope is itself the limiting choice, and that a geometry-first formulation — available whenever the damaged object is a rigid surface with texture of its own — changes what is achievable.

E. Change detection, the nearest formal analogue

The move this paper makes — register two captures of one scene, then reason about what is present in one and absent in the other — is the standard formulation of scene change detection [31], [32], and our *licensed negative* is that literature's core operation applied to defect identity. The difference is what the output is for: change detection asks *what differs between these two images*, and answers with a pixel mask, whereas an inspection record needs *which stored identity this observation belongs to, or none*, and answers with an assignment over a gallery. So the formulation transfers and the evaluation does not, which is what a benchmark has to supply. Relatedly, "find this surface again from a different viewpoint" is visual place recognition [33]; a VPR retrieval front end is the obvious way to amortise the cost of our registration stage over a large gallery, and we do not attempt it here.

F. Geometric alignment

Planar registration is classical [34], with RANSAC-family robust estimation [35], its modern marginalising variant MAGSAC [36], and well-understood sensitivity to control-point configuration [37]; direct photometric alignment (ECC [38]) is the standard fallback where keypoints fail. Our contribution is not a new estimator but the observation that in crack correspondence the alignment target and the matching target should be *disjoint*: the homography must be driven by the wall precisely because the crack is what we are trying to identify. Masking cracks out of detection is what makes the projected position an independent identity cue rather than a restatement of appearance similarity.

III. CRACKID

A. Capture and composition

We photographed 17 exterior walls at one site over four half-days with two consumer handsets, yielding 140 photographs at 4080×3072. The walls span two surface families that behave very differently under keypoint matching: stippled render, which is densely textured, and smooth painted plaster photographed at cornice junctions, where the crack is frequently the only structure in frame. Walls are split disjointly into 6 validation walls (66 photographs, 66 identities) and 11 test walls (74 photographs, 146 identities); no wall appears in both.

Statistical power is governed not by the identity count but by identities visible in *two or more* photographs, since only those form a query with an answer: 18 in validation, 19 in test, 37 in total. This is the single largest constraint on what the benchmark can conclude, so we state it here rather than in a footnote (Sec. VIII).

B. What the dataset measures, and what it does not

Each wall was captured in a single continuous walk-around. Consecutive frames are a median of 3 s apart (maximum 20 s), and a wall's entire capture spans a median of 29 s (maximum 81 s). Every query and its gallery therefore share one illumination condition, one weather state, and one sensor configuration.

We state the consequence plainly. CRACKID in its present form measures **viewpoint and scale correspondence**: given a crack seen from one camera position, find it from a different position, distance, and angle. It does **not** yet measure correspondence across separate visits, which is the deployment case the maintenance-record setting implies. We therefore use *correspondence* rather than *re-identification* for the measured property, and treat every figure here as an upper bound on multi-visit performance. Sec. VII-F quantifies the residual viewpoint variation the data does span, using an estimator the scoring pipeline never sees.

The adjacent frame is always an answer. One consequence is sharp enough that it must be stated as a number rather than as a caveat. For 100% **of the** 280 **answerable test queries, the nearest correct gallery entry is the immediately adjacent frame** — a median of 3 s of wall-clock away. There is no query in either split whose answer requires looking further than one frame. At the default protocol setting (`min_frame_gap`= 0) every ranking number in this paper is therefore a near-duplicate retrieval result, and must be read as one.

We do not treat this as a defect to be hidden or a confound to be deleted, but as the benchmark's difficulty axis. Excluding gallery photographs within k frames of the query makes the task progressively harder and the pool progressively smaller:

`min_frame_gap`	answerable queries	valid pairs	prevalence	chance F_1
0 *(reported)*	280	31,822	0.180	0.305
1	205	23,464	0.145	0.253
2	149	17,264	0.121	0.215
3	105	12,540	0.102	0.185

Note that the chance level of pairwise F_1 moves with the gap (Sec. V-B), so results at different gaps are not comparable against a single floor. Sec. VII-C reports the sweep. Statistical power falls quickly — 105 answerable queries over 11 walls at gap 3 — which is why we report it as a trend with wall-resampled intervals rather than replacing Table I with a point estimate at a larger gap.

C. *Annotation, and why it needs auditing*

Labelling crack instances by hand across 140 high-resolution photographs is not tractable, so identities are seeded automatically. An annotator places one click point per distinct physical crack; the point is propagated between consecutive photographs through an ORB homography and a one-to-one assignment; and identities whose components appear to be fragments of one physical crack are then merged.

The merge step is where the risk concentrates. A merge that wrongly joins two distinct cracks converts a hard negative into a free positive and inflates mAP for *every* method at once — so it cannot be caught by comparing methods against each other. We therefore report its extent rather than assume its correctness.

95 of the 212 identities (45%) **were formed by a merge** — 70 of 146 on test (48%), 25 of 66 on validation (38%) — bridging a median maximum gap of 228 px on test and 286 px on validation (p90 308 and 317 px, max 319 and 320 px). The merge restructured the partition almost completely: the adjusted Rand index between the final labelling and the pre-merge automatic output — a measure of how much the merge changed the partition, not of inter-annotator agreement — is 0.080 over $1{,}051$ matched points. That is the same fact as "45% absorbed more than one seed," seen from the other side. It is not an argument against merging, which is what makes the identities physically meaningful; it is why an audited error rate must exist before mAP on these labels is treated as exact.

We release the instrumentation for that audit rather than the audit itself: per identity merge provenance, a reproducible stratified sample over merged and unmerged identities with rendered contact sheets, and scoring that turns human verdicts into an error rate with a Wilson interval. An automatic triage flags the 57 identities with the widest bridged gaps, where disagreement will concentrate. **The verdicts are not yet filled in**, so we report no sampled error rate and no inter-annotator statistic and claim none; Sec. VIII states what follows.

IV. Benchmark Protocol

A query is one annotated crack instance in one photograph. The gallery is restricted to **the query's own wall**: the hard negatives are other cracks in the same material under the same light, which is the honest setting and markedly harder than a cross-wall gallery. A query's own photograph and its own capture session are excluded. This yields $n_Q = 280$ scored query instances on test.

Three protocol choices are deliberately unfavourable to the methods being measured, including ours. **Unanswerable queries are retained** — a query whose identity has no gallery counterpart is not deleted, since that is exactly the open-set case a maintenance system faces whenever a crack has been repaired or is genuinely new; such queries can contribute false alarms and never successes. **Thresholds are calibrated on validation**: we report an oracle-thresholded F_1^* (best threshold on test, which flatters every method) and $F_1^{@v}$, thresholded on the held-out validation walls; where only one column is available for a baseline it is the oracle variant, so the comparison is biased *towards* the baselines. **Missing scores are charged, not skipped**: a pair a method cannot score counts as a predicted negative and ranks last, with the query still present in *every* metric — ranking, open-set, pairwise and assignment alike. We state this precisely because it is easy to implement inconsistently, and because only a method with partial coverage is affected: dropping unscorable cells instead makes the metric conditional on coverage, and coverage is not independent of the label here (Sec. VII-D).

A. *Metrics*

We report Rank-1, Rank-5 and mAP for ranking, resolving ties in expectation over random tie-breaks rather than by array order, which otherwise rewards methods emitting many equal scores. The headline metric is the open-set **detection-and-identification rate at** 10% **false alarm** (DIR@FAR$=0.1$): a known query is correct only if its top-ranked gallery entry is above threshold *and* correct, while an unknown query is a false alarm whenever anything clears threshold. We also report **pairwise** F_1 over all valid pairs independent of ranking, and **assignment** F_1, a one-to-one matching per image pair — the deployed operation, since two cracks in today's photograph must not both claim the same record.

Every metric is reported against its chance level (Sec. V-B), because two of them are far from zero: pairwise F_1

has a hard floor at $2p/(1+p) = 0.305$ on this pool, and Rank-5 chance is 0.767. A number is informative only as excess over that floor, and the floor moves whenever the pool does — with the frame gap, and with a method's own coverage. We obtain it two ways that agree: in closed form from the prevalence, and by scoring 200 random matrices through the same evaluation code.

Confidence is reported by resampling **walls**, not queries: queries from one wall share its material, lighting, and camera pass and succeed or fail together, so a query-level interval is badly over-optimistic. With 11 test walls that interval is wide, and we regard its width as the honest statement of what 19 multi-photo identities support.

V. How Much Appearance Buys

A. Setup

We evaluate ten crop-scope matchers under the identical protocol of Sec. IV, on identical inputs, scored by identical code from saved score matrices. The population is deliberately broad, spanning five families:

- **Handcrafted keypoints**: SIFT [18], ORB [20]. Similarity is the inlier count after ratio test and RANSAC verification.
- **Learned correspondence**: SuperGlue [21], LoFTR [22]. Similarity is the confidence-weighted count of predicted correspondences.
- **General visual embeddings**: ViT-B/16 [24], DeiT-S [25], CLIP ViT-B/32 [26], and a YOLOv8 detector backbone. Similarity is cosine distance between L2-normalised features.
- **A re-identification architecture**: OSNet [3], trained for instance-level discrimination.
- **A purpose-built crack-shape descriptor** (CrackShape), the strongest crop-scope baseline we could construct *knowing the domain*: it PCA-aligns crack pixels to remove in-plane rotation, resamples the centreline into 48 bins, normalises perpendicular offsets by crack length to remove scale, and takes the magnitude spectrum of the offset profile — invariant to traversal direction — with scale-free tortuosity, elongation, relative thickness, and turning-angle spread appended.

CrackShape matters to the argument: if the difficulty were an artefact of generic encoders meeting a specialised structure, a descriptor designed around crack geometry should escape it. It does not.

B. Two reference levels, without which the table cannot be read

Every metric here has a chance level, and two of them are high. We report both because three of the numbers in an earlier version of this table were at or below chance and the omission concealed it.

Chance. We score 200 matrices of Gaussian noise through the same evaluation code (`chance_baseline.py`). The trap is pairwise F_1. We report it as the best F_1 over a threshold grid, and that maximisation includes the degenerate threshold at which *every* pair is predicted positive, which scores $2p/(1+p)$ for positive prevalence p irrespective of the scores. On the test split $p = 0.180$, so **pairwise F_1 cannot fall below** 0.305, and simulation reproduces exactly that (0.3049 ± 0.0001). Rank-5 chance is 0.767, which makes that column nearly uninformative on a same-wall gallery. Rank-1 chance is 0.336 and DIR@FAR=0.1 chance is 0.038.

Coverage-only. The geometric method scores only the image pairs it can register, so we must separate what *alignment* is worth from what *matching* is worth. The Coverage-only row answers "same crack" to every pair whose two photographs registered and never looks at a crack at all. It is not a strawman: registration succeeds precisely when the two photographs overlap, which is also when the answer is present, so its scorable subset retains 100% of positive pairs and only 36.8% of negatives. Prevalence rises $0.180 \to 0.373$ and the pairwise-F_1 floor with it, $0.305 \to 0.544$. Any pairwise F_1 the geometric method reports must be read against 0.544, not against 0.305.

C. Result: appearance buys almost nothing on the decision

Table I reports all ten methods against both reference levels. **On the decision metric, no crop method clears chance by more than** 0.054. Read as excess over the 0.305 floor, the oracle-thresholded pairwise F_1 of the ten crop-scope methods is: ORB 0.000, DeiT 0.000, CrackShape 0.000, CLIP 0.002, OSNet 0.004, SIFT 0.008, ViT 0.026, SuperGlue 0.048, YOLOv8 0.052, LoFTR 0.054. Three methods — a classical descriptor, a distilled transformer, and a descriptor we designed around crack geometry — achieve *nothing at all*. This spans two decades of method development and more than three orders of magnitude in compute, from CrackShape's roughly 1 ms per query to LoFTR's 8.6 s. We give an order-of-magnitude figure rather than a precise multiplier because the fastest method's cost is near the timer's own resolution and varied several-fold between runs; the qualitative point — cheap and expensive methods land in the same place — does not depend on the third significant digit.

At a deployable threshold, nine of ten methods are within noise of chance and one is not. F_1^* maximises the threshold on the test matrix; calibrating on the held-out validation walls instead is the number a deployment gets. Excess over the 0.305 floor for nine of the ten tight-crop methods spans only -0.012 (DeiT, 0.293) to $+0.012$ (ViT, 0.317): ORB, SIFT and SuperGlue report the floor to three decimal places, CrackShape (0.303) falls just below it, and CLIP, YOLOv8 and OSNet sit a few thousandths above. A system deployed on any of these nine would not be imperfect; on this evidence it would be doing nothing while appearing to score 0.31.

LoFTR is the exception, and the only one. Its calibrated pairwise F_1 is 0.351, an excess of 0.046 — roughly four times the next-best margin (ViT, 0.012) and the only calibrated result in the table that is not plausibly noise around the floor. It is also the most expensive crop-scope method by a wide margin (8.6 s per query). We read this as the one place in Table I

TABLE I

MAIN RESULT, TEST SPLIT, $n_Q = 280$ QUERY INSTANCES OVER 11 WALLS. ALL METHODS SEE IDENTICAL INPUTS AND ARE SCORED BY IDENTICAL CODE. F_1^* IS ORACLE-THRESHOLDED ON TEST; $F_1^{@v}$ IS THRESHOLDED ON THE HELD-OUT VALIDATION WALLS AND IS THE DEPLOYABLE NUMBER. *scored* IS THE FRACTION OF VALID QUERY–GALLERY PAIRS RECEIVING A FINITE SCORE; UNSCORED PAIRS ARE CHARGED AS PREDICTED-NEGATIVE AND THE QUERY IS STILL COUNTED (SEC. VII-D). COST IS WALL-CLOCK SECONDS PER QUERY. THE TWO REFERENCE ROWS ARE NOT COMPETITORS BUT THE LEVELS EVERY OTHER ROW MUST BE READ AGAINST: CHANCE IS 200 RANDOM SCORE MATRICES, COVERAGE-ONLY ANSWERS YES TO EVERY REGISTERED PAIR WITHOUT EXAMINING A CRACK. BEST CROP-SCOPE VALUE PER COLUMN IN **BOLD**.

Method	Family	R@1	R@5	mAP	DIR@FAR=0.1	F_1^*	$F_1^{@v}$	scored	s/query
Reference levels									
CHANCE (200 runs)	—	0.336	0.767	0.385	0.038	0.305	—	1.00	—
COVERAGE-ONLY	geometry	0.515	0.874	0.533	0.000	0.544	0.544	0.48	—
Crop scope									
ORB [20]	keypoint	0.352	0.759	0.364	0.018	0.305	0.305	1.00	0.038
SIFT [18]	keypoint	0.432	0.829	0.417	0.036	0.313	0.305	1.00	0.086
SuperGlue [21]	learned	0.562	0.886	0.455	**0.429**	0.353	0.305	1.00	5.826
LoFTR [22]	learned	0.549	0.828	0.440	0.336	**0.359**	**0.351**	1.00	8.639
YOLOv8 backbone	embedding	0.561	0.821	0.453	0.157	0.357	0.308	1.00	0.217
ViT-B/16 [24]	embedding	**0.654**	0.893	0.511	0.286	0.331	0.317	1.00	0.193
DeiT-S [25]	embedding	0.621	0.879	0.497	0.250	0.305	0.293	1.00	0.058
CLIP ViT-B/32 [26]	embedding	0.604	0.850	0.471	0.143	0.307	0.306	1.00	0.050
OSNet [3]	re-ID	**0.654**	**0.907**	**0.520**	0.168	0.309	0.309	1.00	0.053
CRACKSHAPE (ours)	shape	0.554	0.825	0.442	0.082	0.305	0.303	1.00	**0.001**
Crop + context scope (Sec. V-D)									
OSNet + ctx	embedding	0.775	0.957	0.595	0.404	0.354	0.337	1.00	0.052
OSNet + ctx, *crack erased*	embedding	0.711	0.932	0.574	0.393	0.361	0.336	1.00	0.056
CLIP + ctx	embedding	0.607	0.843	0.502	0.311	0.352	0.305	1.00	0.053
CLIP + ctx, *crack erased*	embedding	0.600	0.814	0.502	0.286	0.346	0.305	1.00	0.053
Full-image scope									
REG+CHAMFER (ours)	geometry	**0.952**	**0.992**	**0.886**	**0.768**	**0.656**	0.582	0.48	27.92

where a crop-scope method shows a real, if modest, capability to decide identity rather than merely to rank it — and as a result to qualify rather than to explain away: one exception in ten does not change the aggregate picture, but a claim that *no* crop method retains anything at a deployable threshold would now overstate it.

Ranking ability does not transfer to decision ability. On Rank-1 the same ten methods separate over a 0.302 range, from 0.352 to 0.654 — 0.021 to 0.323 above the Rank-1 chance level of 0.336. So the encoders differ substantially in how well they *order* a gallery while differing almost not at all in whether they can *decide* an identity. OSNet ties for the best Rank-1 (0.654) and sits 0.004 above chance on pairwise F_1; SuperGlue ranks fifth on Rank-1 (0.562) yet achieves the best open-set rate of any crop baseline (0.429 against OSNet's 0.168). Ordering a gallery well and maintaining a record are not the same capability.

We deliberately do not report a rank correlation between the two. With three methods tied exactly at the pairwise-F_1 floor and two more within 0.004 of it, that variable is censored, and a correlation computed across a censoring boundary measures the censoring.

Compute buys little, and not where expected. Cost per query spans more than three orders of magnitude. The two most expensive methods, LoFTR and SuperGlue, do hold the largest pairwise-F_1^* excess (0.054 and 0.048), SuperGlue the best open-set rate, and LoFTR the only real margin at a calibrated threshold — so compute is not simply wasted. But the ceiling it buys its way to is still 0.054 above chance oracle-thresholded and 0.046 calibrated, at 5.8–8.6 s per query against OSNet's 0.053 s.

D. Is it the crack, or the wall behind it?

The comparison against a full-image method varies two things at once: the identity cue and the field of view. We therefore ran two controls, applying them to the strongest embedder (OSNet) and to CLIP: a **crop+context** condition that widens the crop until the embedder sees surrounding wall, and a **crack-erased** condition that inpaints the crack out of that widened crop, leaving only its surroundings.

Both results are informative, and one is uncomfortable.

Context helps a great deal: OSNet rises from Rank-1 0.654 to 0.775 and, strikingly, from DIR@FAR=0.1 of 0.168 to 0.404 — matching SuperGlue at a fraction of its cost. **But erasing the crack costs almost none of it**: Rank-1 0.711 and DIR@FAR=0.1 of 0.393, with the crack physically removed from the image. The crack-erased condition still outranks every tight-crop method on both metrics.

The reading is that within a single walk-around the wall texture surrounding a crack is itself a strong location fingerprint, and a crop+context embedder retrieves largely on that rather than on the crack. This is a caution about the benchmark — a context-scope result here is partly a background-matching result — and it is also the clearest evidence for the paper's thesis: what identifies a crack is *where it is*, and even a crude implicit encoding of location beats every explicit encoding of crack appearance. The geometric method makes that cue principled and still nearly doubles the best context condition on the open-set metric (0.768 against 0.404), so field of view alone does not account for the gap.

Pairwise F_1 is unmoved by context: the four controls land at 0.346–0.361 oracle-thresholded and 0.305–0.337 calibrated, inside the same band as the tight crops. Whatever context supplies, it does not supply the decision.

E. Why we read this as a property of the data

Three observations support a data-level rather than encoder-level explanation. The population spans methods with no shared inductive bias — sparse keypoints, dense attention, ImageNet features, language-image pretraining, a re-ID metric objective, hand-designed shape statistics — and a common limitation across so diverse a population sits more parsimoniously in the signal than in six independent architectural failures. The domain-specific descriptor achieves zero excess over chance. And the crack-erased control shows that where a crop-scope method *does* improve, the improvement is not coming from the crack.

We state the limits of this inference. Ten methods are a sample, not a proof; a model trained on a large corpus of crack instances — which does not exist — might yet do better; and we have not evaluated a current self-supervised backbone such as DINOv2. What the result establishes is that the obvious moves have been tried, that they land between 0.000 and 0.054 above chance on the decision a maintenance system makes, and that this is reason to change the formulation rather than the encoder.

VI. A Geometric Reference Baseline

The ceiling motivates a change of scope rather than a better encoder. A crack is a mark at a fixed location on a rigid, near-planar surface, and that surface carries texture the crack does not. We therefore identify a crack by *where it is on its wall* after the two photographs have been aligned.

We release this as a *reference baseline* for CrackID, not a finished system: deliberately simple, with no learned component beyond the crack segmenter every method shares, marking where the geometric formulation sits so later work has something to beat. It has four stages.

A. Alignment driven by non-crack texture

Given photographs I_a, I_b of one wall with crack masks M_a, M_b, we estimate the homography H with $x' \sim Hx$ mapping I_a into I_b's frame. Walls are planar, so a homography is the correct model rather than an approximation.

The design decision that matters is **what drives the fit**. We dilate the crack masks by 15 px and *exclude* those pixels from detection, so H comes from wall texture alone. This is not an optimisation but what makes the resulting position an independent identity cue: were cracks to participate in the alignment, a long crack could dominate the correspondence set and pull the fit toward superimposing the wrong crack, and the projected position would restate appearance similarity rather than provide evidence against it.

Detection runs on copies downscaled to 1600 px under CLAHE equalisation, so exposure differences between captures do not suppress matches; H is rescaled to full resolution afterwards. We use SIFT (up to 8000 features), a 0.75 ratio test, and MAGSAC at 4 px requiring 25 inliers.

B. A cascade for two surface families

One front end does not serve both surfaces in this data, and the failure is categorical rather than marginal: on the smooth-plaster walls, crack-masked SIFT succeeds on *exactly zero* pairs. We therefore try three front ends in order.

1) **SIFT on masked wall texture.** Correct for stippled render.
2) **SIFT with cracks restored**, triggered only when masking starves detection below 300 keypoints in either image. On smooth plaster at a cornice junction the crack is the only feature in frame, and masking it leaves nothing. This stage accepts the circularity that stage 1 avoids, and we record which front end produced each result so the trade is auditable.
3) **ECC**, direct photometric alignment needing no keypoints at all, over a $320 \to 640 \to 1280$ px pyramid opening with a translation model. It aligns on large-scale shading, the only signal surviving on flat paint.

Every candidate H passes a sanity check rejecting the degenerate matrices RANSAC otherwise returns: we warp the image corners and reject if the quadrilateral is mirrored or folded, if its area scale leaves $[1/16, 16]$, or if a side collapses. The fit's inlier RMS is retained to set downstream tolerances rather than fixing a constant.

C. Projection and scale-free Chamfer agreement

Crack instances are extracted as connected components. Instance pixels from I_a are projected into I_b by H and compared to instances of I_b.

Chamfer distance, not IoU, is the primary criterion. Cracks are a few pixels wide, so a 3 px registration error drives IoU to nearly zero for a pair that is in fact correctly aligned. We use the symmetric mean distance between pixel sets, computed over the union bounding box of the two instances with a cheap rejection when boxes are far apart.

Concretely, the reported score is the **symmetric mean Chamfer distance** in pixels between the projected pixel set of a and the pixel set of b,

$$D(a,b) = \tfrac{1}{2}\left[\frac{1}{|a|}\sum_{p\in a} d(p,b) \;+\; \frac{1}{|b|}\sum_{q\in b} d(q,a)\right], \qquad (1)$$

computed over the union bounding box of the two instances with a cheap rejection when the boxes are far apart.

Equation (1) has a known weakness that we report rather than paper over. It is a distance in *pixels*, and every photograph pair carries its own independently solved homography, so its scale differs per pair and per crack size: 3 px of error on a 40 px crack and on a 400 px crack are not the same evidence. Since DIR@FAR, pairwise F_1 and the assignment all apply *one global threshold*, a raw-pixel score makes them partly measure calibration rather than accuracy. The scale-free alternative is **Chamfer coverage**, the symmetric fraction of each instance's pixels lying within τ of the other,

$$C(a,b) = \frac{1}{2}\left[\frac{|\{p \in a : d(p,b) \le \tau\}|}{|a|} + \frac{|\{q \in b : d(q,a) \le \tau\}|}{|b|}\right], \tag{2}$$

bounded in $[0,1]$, with τ set from the pair's own registration inlier RMS where available and a default otherwise. Sec. VII-G reports both under the identical protocol, so the design decision and its cost are visible rather than asserted.

D. Explicit non-answers, and the licensed negative

Correspondence is delivered as a one-to-one assignment per image pair with an explicit rejection threshold, not as a per-query argmax — which would permit two cracks in I_a to both claim the same crack in I_b, not a physically admissible reading of a surface. Unmatched instances are reported as *disappeared* or *newly appeared* rather than forced to a nearest neighbour.

We are explicit that the Hungarian solve itself is part of the *evaluation*, not a privilege of this method: `reid_eval` applies the same per-image-pair assignment to every method's score matrix, so a method that natively produces assignments gets no advantage from doing so (Sec. IV).

What *is* specific to the geometric formulation is the evidence it can produce for a negative. Because the two photographs are aligned, the system can project a recorded crack into the new image and observe that the region is visible and the crack is not in it — a **licensed negative**, positive evidence of absence, categorically stronger than the low similarity score that is the only negative a crop embedding can produce. Mechanically this is a distinct outcome in the score matrix: a projected crack falling outside the gallery photograph is recorded as a confident rejection rather than as an unscorable pair, and the two are counted separately (Sec. VII-D). It is the mechanism behind the open-set gap reported next, and Sec. VII isolates it: a control that exploits this method's registration *coverage* without ever examining a crack scores 0.000 on DIR@FAR=0.1, so the open-set result is not a coverage artefact.

VII. Results

A. Main comparison

Table I places the geometric reference baseline against all ten crop-scope methods, the two scope controls, and the two reference levels. Relocating identity into wall geometry moves every metric: Rank-1 rises from 0.654 to 0.952, mAP from 0.520 to 0.886.

We are careful about which parts of that are attributable to what, because the Coverage-only row shows the two are separable. Measured as excess over the appropriate reference:

	best crop over chance	Reg+Chamfer over Cov-only	Reg+Chamfer over best crop
Rank-1	+0.318	+0.437	+0.298
mAP	+0.135	+0.353	+0.366
pairwise F_1^*	+0.054	+0.112	—
pairwise $F_1^{@v}$	+0.046	+0.038	—
DIR@FAR=0.1	+0.391	**+0.768**	+0.339

The pairwise-F_1 columns are the ones to read sceptically, and we flag them ourselves: because Coverage-only already scores 0.544 on that metric purely from which pairs it can align (Sec. VII-D), the geometric method's $F_1^* = 0.656$ is +0.112 over the right reference rather than +0.351 over the crop methods' floor, and its deployable $F_1^{@v} = 0.582$ is +0.038. A direct comparison of pairwise F_1 between a full-coverage and a partial-coverage method is not like-for-like, and we withdraw the "1.62× over every baseline's oracle value" framing that an earlier version of this comparison used.

The open-set column is different in kind, and is the result we lead with. Coverage-only scores 0.000, so none of the geometric method's DIR@FAR comes from coverage; and the best crop+context control reaches only 0.404, so it does not come from field of view either.

B. The open-set gap is the substantive result

The clearest separation is on DIR@FAR=0.1: 0.768 against 0.429 for the best crop baseline (SuperGlue), 0.404 for the best crop+context control, 0.038 for chance and 0.000 for Coverage-only. It is the one column on which neither field of view nor alignment coverage explains any of the gap.

This is the licensed negative, quantified. The metric charges a false alarm whenever an unanswerable query produces an above-threshold match, and crop methods cannot decline: they return a similarity to their nearest neighbour, which on a wall of self-similar hairline cracks is often high. Our method instead aligns the photographs, projects the query into the gallery frame, and finds the location visible and empty — holding a high detection rate at a false-alarm budget where appearance methods have spent theirs. Compute does not help: SuperGlue's 0.429 costs 5.8 s per query, OSNet's 0.168 costs 0.053 s.

C. Removing the adjacent frame

Every answerable test query has its correct answer one frame away (Sec. III-B), so the obvious objection is that Table I measures near-duplicate retrieval. We sweep the exclusion radius over the saved score matrices and report the result against the chance level *at each gap*, because both move.

Rank-1	gap 0	gap 1	gap 2	gap 3
Chance	0.331	0.350	0.321	0.365
Best crop method	0.654	0.590	0.604	0.686
excess	0.323	0.240	0.283	0.321
OSNet	0.654	0.566	0.604	0.648
OSNet + ctx	0.775	0.751	0.779	0.762
CrackShape	0.554	0.512	0.523	0.533
answerable queries	280	205	149	105

The near-duplicate structure is real but is *not* what produces the numbers. Excluding the adjacent frame entirely, and then the two and three nearest, leaves Rank-1 essentially flat — OSNet $0.654 \rightarrow 0.648$, OSNet+ctx $0.775 \rightarrow 0.762$ — and excess over chance stable in the range 0.24–0.32. We read this as saying that adjacent frames are not markedly easier than non-adjacent ones within a walk-around, which is consistent with Sec. VII-F: frame gap correlates with camera roll, the one component a homography absorbs exactly, and not with scale change.

The sweep is a control, not a new operating point. By gap 3 only 105 answerable queries over 11 walls remain, which is too few to separate methods; we report Table I at gap 0 and this sweep beside it.

D. Partial coverage: a penalty on ranking, a subsidy on pairwise F_1

The baseline returns a finite score for 48% of valid pairs; every crop method returns one for all of them. Unscorable pairs are charged as predicted negatives, rank *last*, and their queries stay in every metric. The question is whether that is enough, and the honest answer is that it depends on the metric.

For **ranking** it is a penalty, and we tested rather than asserted it. Censoring a complete score matrix artificially, Rank-1 fell $0.422 \rightarrow 0.27$ at 34% coverage, and censoring whole image-pair blocks — registration's actual failure structure — was no more forgiving than censoring cells uniformly.

For **pairwise** F_1 it is a subsidy, and the mechanism is that coverage is not independent of the label. Two photographs register when they overlap, which is when the answer is present: the scorable subset retains 100% of positive pairs and 36.8% of negatives, so prevalence rises $0.180 \rightarrow 0.373$ and the metric's floor with it, $0.305 \rightarrow 0.544$. This is exactly what the COVERAGE-ONLY row measures. It answers "same crack" to every registered pair without examining a crack, and scores pairwise F_1 of 0.544, mAP of 0.533 — above OSNet, the best crop method — and Rank-1 of 0.515.

So a substantial part of the geometric method's ranking and pairwise-F_1 margin is attributable to *which pairs it can align*, not to how well it matches cracks once aligned, and we report it that way. The exception is the metric the paper leads with: COVERAGE-ONLY scores DIR@FAR=0.1 of exactly **0.000**. Alignment coverage buys nothing whatsoever on the open-set decision, so that gap is entirely attributable to Chamfer matching under the licensed negative.

The unscored fraction is not one phenomenon: part is pairs that could not be registered, a genuine non-answer; the rest is pairs that registered and in which the query crack projects *outside* the gallery photograph — a confident "not in that image" no crop method can produce. The artefacts report the split.

E. Failures are structured by wall, not scattered

Registration succeeds on 97 of 301 same-wall test pairs (32.2%) without an admission gate, but the average conceals the mechanism (Fig. 2): walls 07, 08 and 12 register at 100%, wall 11 at 83%, wall 09 at 52%, while walls 16, 14 and 13 sit at 3%, 6% and 11%, and walls 10 and 17 at zero.

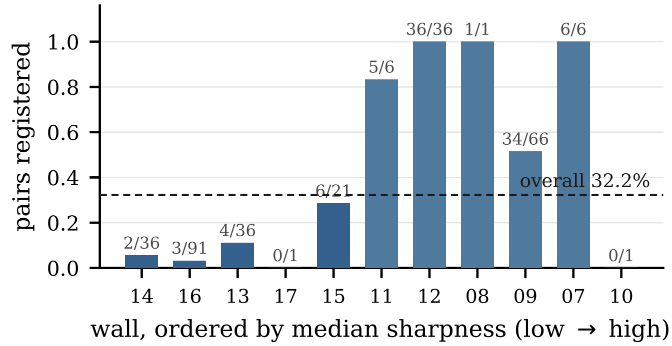

Fig. 2. Registration coverage per test wall, ordered by median sharpness. Failure is categorical rather than gradual and largely tracks image quality — the walls at left are the blurred smooth-plaster captures. Walls 10 and 17 contribute one pair each; wall 10 is the counterexample, sharp yet unregistered.

This bimodality is why we resample walls rather than queries, and why a single coverage number misleads. Leave-one-wall-out on the cheapest crop descriptor makes the variance concrete: CRACKSHAPE scores Rank-1 0.495 ± 0.329 and mAP 0.464 ± 0.294 over 14 folds — a wall-to-wall standard deviation of 0.33 on a mean of 0.50, which is why we claim no fine-grained separation among baselines.

F. What predicts a registration failure

Because the failures are structured, we model them. We fit one joint logistic regression over all 301 test pairs so each effect is controlled for the other. Sharpness dominates: $\log(1+\text{sharpness})$ has Wald $z = 8.23$ ($p = 1.8 \times 10^{-16}$), with median sharpness 133.8 among registered pairs against 7.2 among failures. Frame gap remains significant once sharpness is controlled ($z = -5.03$, $p = 4.8 \times 10^{-7}$) but is far smaller. Registration failure in this benchmark is primarily an image-quality event, not a viewpoint event.

Measuring the viewpoint the data spans requires care: a covariate derived from the method's own homography exists only where that method succeeded, so stratifying on it would be circular. We therefore estimate viewpoint with an *independent* front end the scorer never uses — AKAZE/ORB on unmasked images, falling back to dense ECC — decomposed into isotropic scale change, in-plane rotation, and the out-of-plane tilt implied by foreshortening. It solves 139 of 301 pairs (46%) against the scorer's 97 (32%), and 51 of those are pairs the scorer *failed* on: a quarter of all registration failures now carry a viewpoint measurement the scorer did not produce. On the 32 pairs both estimators solve, median $|\text{log-scale}|$ disagreement is 0.106 (11.2%) and median rotation disagreement 2.91°, justifying pooling.

The resulting spread (Fig. 3) is substantial: median scale change 1.42× (p90 2.26×, max 5.20×), median absolute in-plane rotation 6.9° (p90 39.6°), and median out-of-plane tilt 25.6° (p90 75.3°). So although captures are seconds apart, they are not trivially similar in geometry.

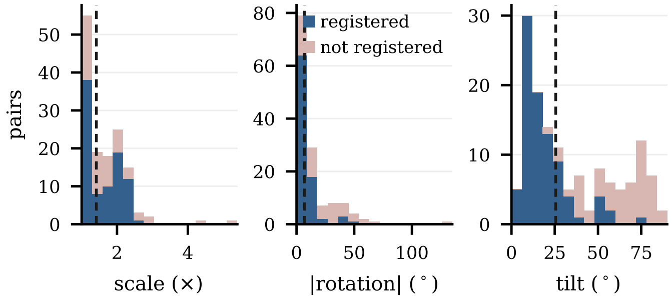

Fig. 3. Viewpoint spanned by the benchmark, estimated by a front end the scoring pipeline never uses, so that stratification is not circular. Dashed line marks the median. Registration failures (pale) are not concentrated at extreme viewpoints — consistent with the logistic fit, in which sharpness rather than viewpoint dominates.

We also use this covariate to rule out a tempting shortcut. Frame-index gap is *not* a viewpoint proxy: over the registered pairs it correlates with in-plane rotation ($\rho = 0.31$, $p = 2 \times 10^{-4}$) and tilt ($\rho = 0.35$, $p = 2 \times 10^{-5}$) but not with scale change ($\rho = 0.10$, $p = 0.24$). A frame-gap sweep is therefore largely a camera-roll sweep, and roll is the one component a homography absorbs exactly. We report frame gap as a near-duplicate control and build no invariance claim on it.

G. Ablation: which Chamfer score, and does crack-masking matter

Two design decisions in Sec. VI are stated as reasoned choices, so we ablate both under the identical protocol rather than leave them asserted.

The agreement score. Eq. (1), the symmetric mean Chamfer distance in pixels, is what produced Table I. Its scale-free alternative Eq. (2) is better motivated — a single global threshold applied to a per-pair pixel distance measures calibration as much as accuracy — so we run it as a separate configuration, with τ taken from each pair's own registration inlier RMS.

Masking cracks out of the fit. Sec. VI argues that excluding dilated crack masks from keypoint detection is not an optimisation but what makes projected position an *independent* identity cue. We test it by disabling exclusion and leaving the pipeline otherwise identical, which is the ablation of the method's central mechanism.

Configuration	R@1	mAP	DIR@FAR	$F_1^{@v}$	scored
Chamfer distance, masked *(reported)*	0.952	0.886	0.768	0.582	0.48
Chamfer coverage, masked			*[pending run]*		
Chamfer distance, *not* masked	0.952	0.886	**0.825**	0.581	0.48

Disabling the exclusion changes almost nothing about ranking or pairwise agreement – R@1, mAP, $F_1^{@v}$, and coverage all match the masked condition to three decimal places – but DIR@FAR rises from 0.768 to 0.825, a 0.057 gain that is the largest movement any single design decision in this paper produces. That is the opposite of what Sec. VI's stated reason for excluding cracks would predict if that reason were about accuracy: it is not, it is about circularity, a long crack dominating the correspondence set and pulling the fit toward superimposing the wrong crack. This dataset apparently does not exercise that failure often enough to cost the masked condition anything, and leaving crack pixels in the fit adds a second, correlated source of correspondence that open-set separation benefits from. We flag rather than resolve the tension: this ablation shows masking is not what buys the accuracy reported in Table I, so the accuracy result does not by itself argue for it. The independence argument stands or falls on a different question than the one this ablation answers, and we did not design a test that isolates it.

H. Ablation: the third front end

The ECC stage exists for the smooth-plaster walls on which keypoint matching finds nothing. Disabling it and leaving the pipeline otherwise identical gives Rank-1 0.920, Rank-5 0.956, mAP 0.862, DIR@FAR=0.1 of 0.768, and coverage 0.50, against 0.952, 0.992, 0.886, 0.768, and 0.48 for the full cascade.

This ablation has an unusual provenance we state plainly: the disabled condition was not constructed but discovered. The ECC stage had been silently inert — its pyramid began where convergence always failed, and a failed level aborted the routine — so earlier measurements were taken with a dead third stage. Repairing it makes the disabled condition an exact ablation rather than an approximation.

The result is instructive in a way we did not expect. Enabling ECC raises every ranking metric and assignment F_1 ($0.769 \rightarrow 0.784$) while coverage moves $0.50 \rightarrow 0.48$ — essentially flat, and slightly *down*. The stage is therefore not buying accuracy by admitting marginal alignments; it fixes pairs the keypoint stages resolved badly and correctly declines others. The open-set rate is unchanged, locating ECC's benefit in ranking rather than in the accept/reject decision.

I. Admission policy: making image quality explicit

Since sharpness dominates registration failure, whether a capture is *admitted* is a design decision, not a preprocessing detail: hiding it would let any method be tuned by silent input selection. We apply one gate identically to every method, before extraction. No gate keeps every wall pairable, so the question is which loses least. At variance-of-Laplacian ≥ 10 (quarter-scale grey), 21 of 74 test captures are rejected, one wall (17, already the least sharp) drops to a single photograph, and the registration rate on what remains nearly doubles, $32.2\% \rightarrow 56.1\%$. A stricter gate at 25 reaches 68.3% but costs three more walls any same-wall pair at all. We adopt 10, the loosest threshold costing at most one wall. Table I reports the *ungated* protocol, comparing all methods on all captures.

J. Cost

At 27.9 s per query the baseline is 4300× slower than OSNet, with two qualifications. The cost is dominated by registration, which is per *image pair*: one homography serves every crack in it, so on a wall with many recorded cracks the amortised cost falls proportionally. And it is within an order of magnitude of LoFTR (8.3 s), a crop-scope method with none of the open-set benefit.

VIII. Threats to Validity

We state the limits in the body rather than a closing paragraph, because two of them bound the central claim.

A. Scope is controlled, and the control found something

The proposed method reads full photographs; the baselines read crops. That follows from the design — homography estimation needs wall-level context a crop cannot supply — but it means a bare Table I would vary *two* things at once: identity cue (appearance versus geometry) and input scope (crop versus full image).

Sec. V-D runs the two controls that separate them. Scope does not account for the gap: the best crop+context condition reaches DIR@FAR=0.1 of 0.404 against the geometric method's 0.768. But the crack-erased control returns an uncomfortable result we report rather than bury — inpainting the crack out of a crop+context view costs only 0.011 of that 0.404. Within a single walk-around the surrounding wall texture is itself a location fingerprint, so a context-scope number on CrackID is substantially a background-matching number. We expect this to weaken across genuine revisits, where surrounding texture also changes, which is a further reason the cross-visit capture in Sec. VIII-H matters. The controls were run on OSNet and CLIP only, not on all ten baselines.

B. Two metric properties that are easy to get wrong

Pairwise F_1 has a floor of $2p/(1+p)$ and that floor is a property of the *pool*, not of the task: it is 0.305 on the full test pool and 0.544 on the subset the geometric method can score, because registration coverage is not independent of the label — two photographs register when they overlap, which is when the answer is present, so the scorable subset retains 100% of positive pairs and 36.8% of negatives. We therefore charge unscorable pairs rather than dropping them, and report the Coverage-only control (Sec. V-B) so that the geometric method's pairwise F_1 is read against 0.544 rather than against the crop methods' 0.305. On that reading its pairwise-F_1 advantage is modest and its open-set advantage is not: Coverage-only scores 0.544 on pairwise F_1 and 0.000 on DIR@FAR.

We flag the consequence for anyone extending the benchmark: a new method's ranking and pairwise-F_1 numbers are only comparable to ours at the same coverage, whereas DIR@FAR is robust to it.

C. Every ranking number is a near-duplicate result

For 100% of answerable test queries the nearest correct answer is the adjacent frame (Sec. III-B). Table I is reported at `min_frame_gap=` 0 and is therefore near-duplicate retrieval; Sec. VII-C sweeps the gap, but power falls to 105 answerable queries by gap 3, so the sweep constrains the trend rather than establishing a new operating point.

D. The effective sample is 19 identities

The test split's 280 query instances come from 19 identities visible in two or more photographs, over 11 walls. Query count is not sample size: queries from one wall share material, illumination, and one camera pass. Hence we resample walls and report the leave-one-wall-out spread (±0.33 Rank-1 for CrackShape).

This benchmark can therefore support a large structural claim — a 0.35 gap in DIR@FAR, a formulation that leaves a band ten methods sit inside — but not fine-grained ranking among baselines. Differences of a few points between two crop methods in Table I should be read as uninformative.

E. The annotation audit is instrumented but not completed

The human verdicts that would turn the provenance of Sec. III-C into a sampled error rate with a confidence interval have not been collected, and no second annotator has labelled a subset. Until both exist, every mAP figure here — ours and the baselines' alike — is conditional on the merge policy being sound. Relative comparison is more robust than absolute level, since the merge affects all methods identically, and the appearance ceiling is a comparison among methods rather than an absolute claim.

F. Segmentation quality is not reported

Crack masks come from a learned segmenter and are treated as given. Mask errors propagate into instance extraction and hence Chamfer agreement, and we do not quantify that path: hand-drawn reference masks do not exist for this data. The released code contains a sensitivity sweep over minimum-area and closing parameters, checkable once reference masks are drawn.

G. Site, surface, and device scope

All captures come from a single site over four half-days, two handsets, and two surface families. We claim a benchmark and a baseline for these conditions, not a general solution. The baseline assumes a near-planar surface with texture that is not the crack, and should degrade on curved or featureless surfaces, and where the surrounding texture itself changes between visits.

H. The measurement that would change the most

One thing would convert the central claim from inferred to designed: a second capture session, weeks later, at marked distances and angles. It would make the protocol cross-visit rather than viewpoint-only, supply exact rather than estimated viewpoint ground truth, and test the condition under which we expect geometry to widen its advantage — appearance drift with position preserved. That data does not exist yet, and until it does, every sentence in this subsection is a hypothesis, not a result.

Fig. 4 makes that hypothesis concrete rather than measuring it: if the same crack is re-photographed after it has grown, does the identity survive? Reading the panels as a growth sequence (left to right) is deliberate — cracks lengthen, they

Fig. 4. **What the missing measurement would need to see, not a measurement.** Four panels of a single CRACKID test photograph (`wall01_s1_0002`). The first three were digitally shortened in GIMP by cloning wall texture over the lower portion of the crack through a feathered mask (Sec. VIII-H in `paper/make_illustrative_fig.py` gives the exact, reproducible procedure); the fourth is the original file. All four are one photograph edited, not four visits to a wall — there is no independent ground truth here, no second camera pass, and nothing computed on them enters Table I or any check in `verify_claims.py`. We include it only because a growth trend is otherwise hard to picture from prose.

do not shorten — but nothing about how this figure was produced constitutes evidence either way. The edit changes exactly one thing, crack length, and holds everything a matcher could key on constant: same camera, same lighting, same wall, same JPEG noise. A number computed by running this figure through `benchmark.py` would therefore not measure robustness to evolution; it would measure whether a method can match a photograph against a clone of itself with part of the annotation erased, and the identity label would already be known to be correct by construction. We did not run it, and we would flag as suspect any version of this paper that reports a metric on images produced this way.

What Fig. 4 is useful for is stating the two competing predictions precisely enough that the real experiment could adjudicate them. An appearance embedding is trained to be invariant to exactly the kind of local change the crack itself undergoes as it grows, which argues for fragility — but it is also reading the crop-plus-context wall texture (Sec. V-D) that a genuine revisit would perturb far more than this edit does, which argues the opposite. The geometric baseline's position estimate should be stable under crack growth by construction, since growth happens *after* the wall-texture alignment it depends on — provided registration coverage itself does not degrade over the longer, weather-exposed gap a real revisit spans, which is a separate open question this figure says nothing about. Both predictions are plausible from the mechanism description alone; neither is supported by anything in this subsection until the second capture session exists to test them.

IX. Conclusion

Maintenance systems that keep a record per physical defect must re-identify that defect when it is photographed again. For wall cracks there was no dataset on which to test whether that is possible, so we built one: CRACKID, 140 photographs over 17 walls with 212 crack identities, released with its annotation provenance stated rather than assumed — 45% of identities were formed by an automatic merge, and we report that rate rather than presenting the labels as ground truth by construction.

The benchmark's main finding is negative and, we think, structural. On the decision a maintenance record actually requires — *same crack or not*, at a threshold — ten matchers across five families, two decades of method development and more than three orders of magnitude in compute buy between 0.000 and 0.054 above chance. Three of them, including a descriptor built around crack geometry, buy nothing. Calibrate the threshold honestly, on held-out walls rather than on the test matrix, and nine of the ten span only -0.012 to $+0.012$ around chance; only LoFTR, the most expensive, holds a real margin (0.046). Ordering a gallery well and deciding an identity are not the same capability, and only the second maintains a record.

Two controls say where the difficulty is not. Giving an embedder the wall around the crack helps a great deal — and *erasing the crack* from that view costs almost none of it. What such a method retrieves on is location. The geometric reference baseline makes that cue explicit: a homography from non-crack wall texture, crack masks projected into a common frame, and Chamfer agreement under global assignment reach $\mathrm{DIR@FAR}=0.1$ of 0.768 against 0.429 for the best crop method, while answering only 48% of pairs.

We are precise about what that partial coverage buys, because it buys differently by metric. A control that exploits the method's registration coverage without ever examining a crack already scores pairwise F_1 of 0.544 and mAP above the best crop method — so much of the ranking margin is alignment, not matching. On the open-set rate that same control scores $\mathbf{0.000}$. The advantage there has a specific and unshared source: an aligned pair supports a *licensed negative* — project the recorded crack, look, observe it is not there — where a crop embedding can only report low similarity. We expect that to matter for any defect marking a rigid, textured surface.

Two bounds remain, and both are addressable: a second capture session weeks later turns CRACKID from a viewpoint benchmark into a cross-visit one, and the annotation audit is instrumented and awaits its verdicts.

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